

IT Administrator

1. What is the process for onboarding a new staff member, who initiates it, and what data is collected?

The manager should inform us when a new staff member joins, then we will create the new staff member's account and assign them the appropriate role, such as staff or manager.

We would need their basic information such as name, contact information, staff ID, and start date. For teachers, we would also need to indicate the instruments they are qualified to teach.

So the process is that we will first create the account for the new staff member, which includes their name, staff ID, contact information, role, start date, and qualified instruments. Then, when they login to the system, they are able to input their own availability and job preference for the week.

2. What authentication and access control system does the school currently use, or should we build one from scratch?

Currently we don't have an authentication or access control system, other than the encrypted Excel file which requires a password to access the information. Since this new system is going to be web-based, I would expect authentication and role-based access to be included. Teachers, managers, and IT admins should only be granted permissions that are relevant to their role in the system.

As the system will be accessible over the Internet, we would also expect it to have a high level of security to prevent unauthorised users from gaining access to sensitive staff and student information. Appropriate security measures should be implemented, to ensure that only authenticated and authorised users are able to access the system and its data.

3. What should each role be able to see about each other? Can a teacher view their colleagues' schedule or availability?

Managers are the ones responsible for allocating jobs to teachers, so they should be able to see all the teachers' availability, assigned workload, job preferences, qualified instruments, etc.

Teachers mainly only need to see their own weekly assignments, monthly workload, availability and job preferences. They should not be able to view each other's schedule or availability.

4. When a staff member leaves the music school, what's the offboarding process — both for their data and their system access?

The manager would inform us, and we would disable the staff member's account so that they are no longer able to login to the system and will not be considered in future rostering and availability selections. Their accounts should be disabled or archived rather than fully deleted since we still need to retain their job assignment history.

5. What happens when a teacher leaves mid-month, or becomes qualified for a second instrument? Who updates the system and what happens to the lessons already assigned to them?

For teachers who leave mid-month, I would expect the manager to inform us so that we can disable the teacher's account on their last day of employment. For any roster changes, the managers should be able to update or re-assign another teacher to a job.

If a teacher becomes qualified to teach another instrument, the manager should inform us so that we can update the teacher's profile, allowing managers to assign them jobs of that instrument.

6. How are physical studio spaces, instrument setups, & facilities be documented & maintained in the system

There should be a maintained list of physical studios, instrument setups, and facilities stored in the system as system resources. Currently there are five studios, but if more are added in the future, we (IT Admins) should be able to update the list by adding, removing, or editing studios.

Each studio should also contain information about the instruments and facilities available in it. Currently only one of our studios has a drum kit, so the system should consider this when the managers assign lessons to studios. However for some instruments like violin, it can be conducted in any studio. Therefore the system should only display a studio as an option if it meets the facility/equipment requirements of the lesson.

If a studio is temporarily unavailable, for example due to renovation or maintenance, we should be able to mark the studio as unavailable or inactive for a specified period. Unavailable studios should not appear as available options when managers are assigning lessons during that period. Once the studio becomes available again, we (IT admins) should be able to reactivate it without having to re-create the studio and its information.

7. Do you get a lot of questions from staff or managers about how to facilitate rostering?

Since we are currently using Excel for the rostering, we don't receive many questions as it is quite straightforward, any changes to the roster would need to be done manually by the staff and managers themselves, which is quite prone to human error.

For the new system, it would be helpful if the rostering process is clear and easy to understand so that users can navigate and use the system without requiring frequent assistance from us. It should also be able to prevent errors made in rostering by automatically checking for conflicts, such as assigning an unavailable teacher or studio or assigning a teacher to an instrument they are not qualified to teach.

Current Infrastructure

8. What devices and operating systems do staff currently use? (School-provided or personal?)

Currently, each staff member is provided with a Thinkpad running Windows. All work communications and administrative work are done through the provided laptop.

9. Does the school have its own servers, or is everything cloud-based?

Most of our existing systems are cloud-based; we do not maintain our own on-premise servers

10. What's the internet connectivity like at the school — reliable Wi-Fi, wired, or spotty?

The internet connection at the school is generally quite reliable and we do not face many connectivity issues. There may occasionally be some downtime if there is an issue with the Internet service provider though.

In the event of a temporary connection loss, we would prefer that the new system could save any information that the user has already entered where possible, so that they do not have to re-enter everything when the connection is restored.

Integration & Data

11. Is there a student management system or enrollment database already in use?

Yes, we do currently have a basic student management system which allows us to maintain student information such as their personal details, contact information, enrolment status, and enrolled instruments.

However, the system is used mainly for maintaining student records and is separate from our current rostering process. Ideally, the new system should be able to retrieve relevant information from this existing database such as their enrolled instrument, enrolment status, and availability.

12. How is teacher data currently stored — spreadsheets, existing software, paper?

We also do have an employee management system where the employment information of all employees are kept. However, this system is mainly used for HR purposes, such as maintaining employee records and employment details.

13. If we need to import historical data (past schedules, student records), where does that data live and in what format?

Our past rosters and scheduling records are mainly stored in Excel files, usually organised by month. The student records are stored in our student management system database.

If historical data needs to be transferred into the new system, we should be able to export the relevant information from these systems and import them into the new system.

14. Are there any APIs or third-party services the school uses that this system might need to connect to?

Currently, we do not have any APIs or third-party services that the new system must connect to. However, since we already have an existing student and employee database in use, it would be useful if the new system could integrate with these systems to reduce the need to manually enter and update the same information across multiple systems.

Security & Access

15. Does the school have existing accounts/credentials for staff (Google Workspace, Microsoft 365, etc.)?

Yes, each staff member is assigned a school email account under our school's domain when they join. These accounts are currently used for work related communications over Microsoft Teams. Teachers' and managers' school email addresses may also be provided to the parents or guardians of enrolled students when necessary, so that they can communicate with the relevant staff regarding lessons, scheduling, or other administrative matters.

16. Are there any data privacy requirements the school must follow (e.g., student data protection)?

Yes definitely, since the system will contain or pull data from the employee and student databases, it should comply with Singapore's PDPA and ensure that all personal information is properly protected and that users can only access information that is necessary for their role.

Support & Maintenance

17. Who currently handles IT support requests from teachers and managers?

We (IT Admins) are the main contact for technical issues such as login problems, account access, or system errors.

18. How do you prefer to receive reports about system errors or issues?

Since we are not a large school, a simple support process should be enough. Staff members could report an issue to IT with their name, description of the error, and possibly a screenshot. It would also be useful to keep a log of reported errors for tracking purposes.

Reliability & Backup

19. What happens if data is accidentally deleted? Is there a backup process in place?

For the new system, we would expect the production data to be backed up automatically at certain intervals, so that if any data is accidentally deleted, we would still have the backup data to fall back on.

Especially for things like staff accounts, schedules and assignments, changes made should also be tracked so that previous information can be recovered if necessary. These records should not be permanently deleted from the database, and should instead be archived or marked as inactive where appropriate.

20. How many people would be using the system at the same time during peak scheduling periods?

We currently only have about 9? active teachers (from the table in the brief, idk if got 9 teachers or is it got teachers can teach two diff instruments)

We currently only have about 9 active teachers, plus the managers and IT/admin users, there would probably be less than 20 active users using the system at the same time during peak scheduling periods.

21. What documents or records does the school currently keep that we should review before designing the system?

The most useful records would be the current staff list and their instrument qualifications, existing teacher schedules, teacher availability records, student enrolment and availability information, studio/room information, current monthly rosters, and teacher workload records.

We also have some rules for rostering which the system should enforce when managers assign jobs, such as the studio constraints, teacher qualifications, opening hours, public holiday closures, and a strict rule that teachers can only teach for a maximum of four continuous hours before having at least a one hour break, after which they can continue with another four hour block.